

# Continuous random variables: density functions

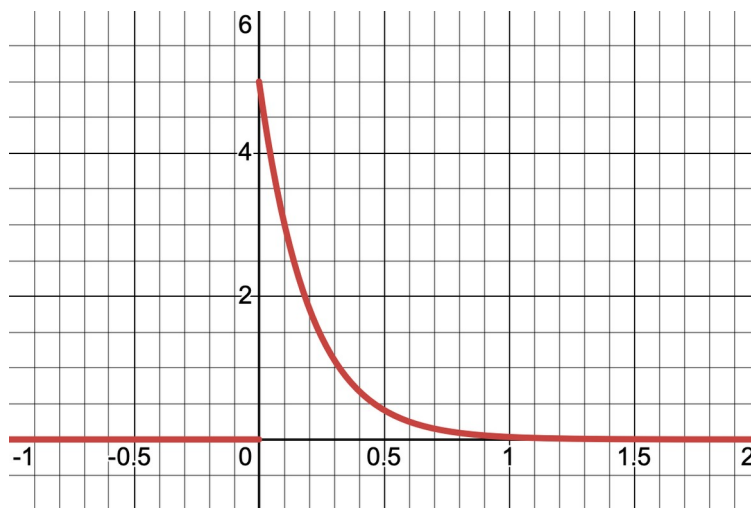
We can model the time gap between when successive e-mail messages arrive in my inbox as a continuous random variable  $X$ . We describe the distribution of the random variable using a “density function”  $f$  defined by  $f(x)=0$ , if  $x<0$  and  $f(x)=5e^{-5x}$ , if  $x\geq 0$ . Here  $x$  is a time interval, measured in hours.

The kind of event that we consider related to  $X$  is  $(a\leq X\leq b)$ , where  $a$  and  $b$  are real numbers. For example, the event  $(0.1\leq X\leq 0.3)$  means that the gap between e-mail messages is between 0.1 hours (6 minutes) and 0.3 hours (18 minutes).

To calculate  $P(a\leq X\leq b)$ , we integrate:

$$P(a\leq X\leq b)=\int_a^b f(x)dx.$$

Here is the graph of  $f$ :



You can think of the density function as the outline of a (density) histogram showing time gaps between many consecutive e-mail messages. When the density is larger, the time gaps are more common.

1. Shade the part of the graph corresponding to the event  $(0.1\leq X\leq 0.3)$ . Calculate  $P(0.1\leq X\leq 0.3)$

2. Calculate  $P(X > 0.5)$ . What does this probability tell you about gaps between my e-mail messages?

3. Calculate  $P(0 \leq X \leq T)$ , treating  $T$  as a variable. What is the limiting value of this probability as  $T \rightarrow \infty$ ? What does the limiting value tell you about gaps between e-mail messages?